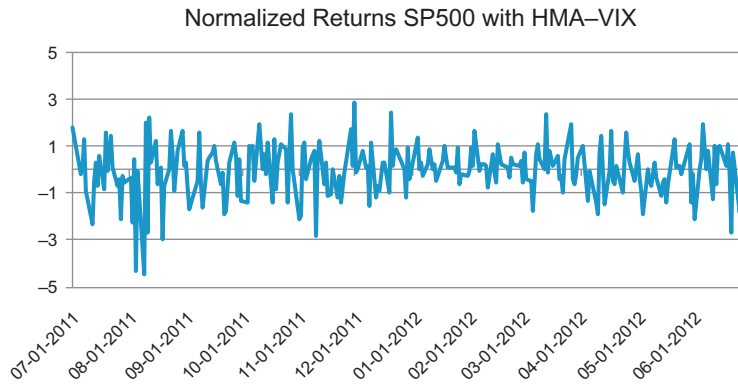




■ **Figure 6.6** Normalized SP500 Returns using HMA-VIX

makes the HMA-VIX model a potentially powerful volatility forecasting model for intraday trading applications and electronic trading algorithms.

Some of the advantages of the HMA-VIX volatility model over other techniques are:

- Reacts to new information sets prior to those events affecting prices. Historical models will only react to new information after it has already affected prices. There is always some degree of lag when using historical models or models based on realized prices.
- Incorporates real-time information from the options market, e.g., forward looking implied volatility, across the full universe of stock. Implied stock volatility is only available for a very limited number of stocks.
- Provides necessary real-time volatility estimates that can be incorporated into trading applications and electronic trading algorithms.
- Allows algorithms to make real-time revision to their execution strategies, limit order model, and smart order routing logic in real-time.
- Performed as well as, and in some cases better than, some of the more traditional volatility forecasting models.

As a follow-up exercise, we propose further research that combines alternative historical volatility measures with forward looking implied volatility terms. For example, combine the GARCH or EWMA models with an implied volatility term. The implied volatility term can be based on the ratio of the VIX index to S&P trailing volatility (as proposed above), or possibly based on the net change or log change in the VIX index from one period to the next. It is often said that execution performance will only be as good as the models that are used to manage the executions, and that those models are only as good as the accuracy of the forecasted explanatory factors. Since